

Question number	Answer	Additional guidance	Mark
6(a)	An answer that includes the following points: - slow twitch fibres are more important to long-distance athletes/10000m runner /marathon runner (1) - they have more mitochondria (1) - and {larger capillary network/ more myoglobin} to supply oxygen (1) - allowing {aerobic respiration/more ATP production} (1)	accept converse accept as the distance run increases so the % of slow twitch muscle fibres increase can run for longer distances without fatigue ignore reference to anaerobic respiration	(3)

Question number	Answer	Additional guidance	Mark
6(b)	Choose an item. sarcomere	accept phonetic spelling	(1)

Question number	Answer	Additional guidance	Mark
6(c)(i)	A calculation showing the following step: - calculation of cardiac output (1) - conversion of cardiac output into $\text{dm}^3 \text{ min}^{-1}$ (1)	$(65 \times 57) = 3705 \text{ cm}^3$ $(65 \times 0.057 \text{ or } 0.065 \times 57)$ 3.71 accept 3.70 accept 3.7 accept 3.705 units not needed as in stem	(2)
	2	1	
	3.71	3705	
	3.705	$0.057 \text{ or } 5.7 \times 10^{-3}$	
	3.7	$0.065 \text{ or } 6.5 \times 10^{-3}$	

Question number	Answer	Additional guidance	Mark
6(c)(ii)	The correct answer is B		(1)

Question number	Answer	Additional guidance	Mark
6(c)(iii)	A calculation showing the following step: Correct calculation of the ratio (1)	$(1.54:2.39) = 1:1.55 \text{ / } 0.64:1$ accept $1:1.6 \setminus 1:1.5$ accept $0.644:1$ accept $0.64 \text{ / } 0.644$ (no need for :1)	(1)

Question number	Answer	Additional guidance	Mark
6(d)(i)	An answer that includes two of the following points: - as the running speed increases the cardiac output increases (1) - at speeds over 10 mph cardiac output {decreases/ levels off }(1)	accept linear relationship (up to 10 mph) accept positive correlation accept at 12mph cardiac output has decreased accept cardiac output decreases after 11mph ignore highest cardiac output at 11mph	(2)

Question number	Answer	Additional guidance	Mark
6(d)(ii)	An explanation that includes three of the following points: - increase in carbon dioxide in blood/ decrease in blood pH (1) - detected by chemoreceptors (1) {medulla /CVC} sends (more) impulses {via sympathetic nervous system/ to SAN} (1) - SAN {depolarises more frequently / causes increased rate of heart muscle contraction} (1)	ignore ref to increased oxygen demand ignore muscles unqualified	(3)

Question number	Answer	Additional guidance	Mark
6(d)(iii)	A description that includes two of the following points: - transported to the liver (1) - converted to {pyruvate /glucose} (in the liver) (1) (pyruvate / glucose) used in {link reaction / Krebs cycle/aerobic respiration} (1)	ignore lactate used in aerobic respiration / link reaction / Krebs cycle no if IT is used in link reaction	(2)